Last name _____

First name _____

LARSON—MATH 310–HOMEWORK WORKSHEET 09
Matrix Inverse Calculations

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Problems

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

Let the columns of matrix A be $\vec{a}_1, \vec{a}_2$.

1. Check that the columns $\vec{a}_1, \vec{a}_2$ are linearly independent (and, thus, that A has an inverse).

Inverse via Gram-Schmidt

2. (a) Apply Gram-Schmidt to $\vec{a}_1, \vec{a}_2$ to find an orthonormal list of vectors $\vec{q}_1, \vec{q}_2$.
   (b) Use these as columns for the Q matrix.
   (c) Check that $Q^T = Q^{-1}$ (by checking tht, $QQ^T = I$).
   (d) Find coefficients $R_{1,1}, R_{2,1}$ so that $\vec{a}_1 = R_{1,1}\vec{q}_1 + R_{2,1}\vec{q}_2$.
   (e) Find coefficients $R_{1,2}, R_{2,2}$ so that $\vec{a}_1 = R_{1,2}\vec{q}_1 + R_{2,2}\vec{q}_2$.
   (f) Let
   $$R = \begin{bmatrix} R_{1,1} & R_{1,2} \\ R_{2,1} & R_{2,2} \end{bmatrix}$$
   (g) Check that $A = QR$. If not, go back and find your error.

We know that $A^{-1} = R^{-1}Q^{-1} = R^{-1}Q^T$. We showed that $A^{-1} = R^{-1}Q^T$. We'll find A^{-1} by finding R^{-1}.

Now we'll find R^{-1} with a direct calculation of its columns. They must satisfy $RR^{-1} = R\begin{bmatrix}\vec{x}_1 & \vec{x}_2\end{bmatrix} = \begin{bmatrix}\vec{e}_1 & \vec{e}_2\end{bmatrix} = I$.

(h) Solve: $R\vec{x}_1 = \vec{e}_1$, which is:

$$R\begin{bmatrix}x_{1,1}\\x_{2,1}\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}.$$

Write the corresponding equations.

(i) Solve your system to find $x_{1,1}$ and $x_{2,1}$.

(j) We need to solve: $R\vec{x}_2 = \vec{e}_2$, which is

$$R\begin{bmatrix}x_{1,2}\\x_{2,2}\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}.$$

Write the corresponding equations.

(k) Solve your system to find $x_{1,2}$ and $x_{2,2}$.

(l) Let

$$R^{-1} = \begin{bmatrix}x_{1,1} & x_{1,2}\\x_{2,1} & x_{2,2}\end{bmatrix}$$

Check that $RR^{-1} = I$.

(m) Find $A^{-1} = R^{-1}Q^T$.

(n) Check that $AA^{-1} = I$.

Inverse via 2×2 Formula

Only in the 2×2 case is there a simple formula. The previous process can be streamlined (in particular there is a slick formula for finding the R matrix coefficients)–and **works for invertible matrices of any size**.

(a) What is the 2×2 formula for the inverse of an invertible matrix A?

(b) Use this formula to find A^{-1}.